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Unit 4 : Session 12 : SLO 2 | Influence on Instruction Sets Fill in the Blanks

I. Fill in the Blanks

1. RISC is a small set of simple, fixed-length instructions.
2. CISC CISC instruction sets allow memory-to-memory operations for compact code.
3. "make the common case fast" (Amdahl's) frequently used instructions should be fast.
4. SIMD (vector) modern processors use SIMD extensions for parallel data processing.
5. code density embedded systems prioritize compact code to minimize memory usage.
6. fixed-length the fixed-length format simplifies pipelining but may reduce code density.
7. Predicated predicated execution conditionally executes instructions, removing branches.
8. Backward backward compatibility forces x86 to keep complex legacy features.
9. load-store the RISC load-store model requires explicit load/store instructions for memory access.
10. Multiple (complex) many addressing modes increase flexibility but complicate hardware design.